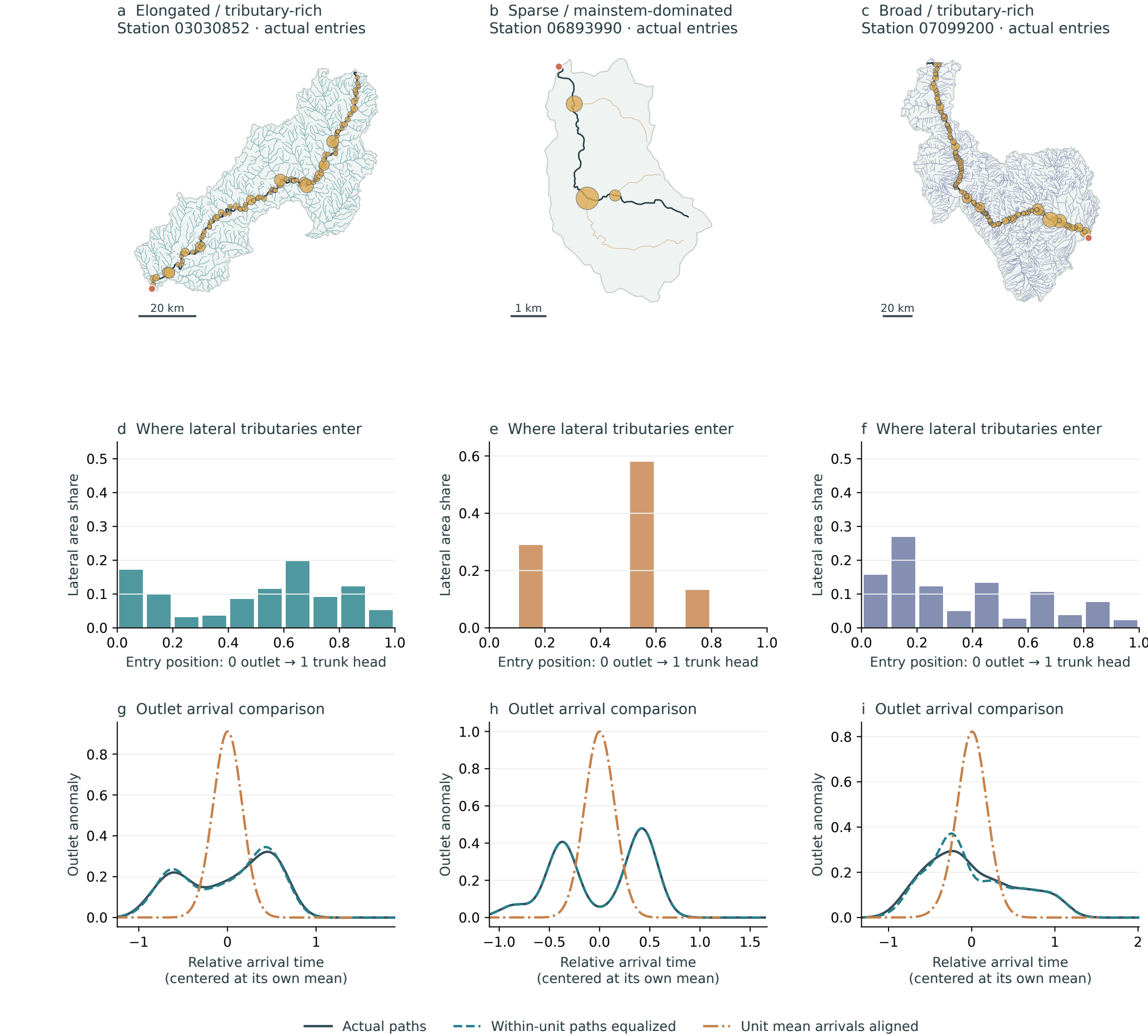


Whole-network tributary layout and outlet signal overlap

Original mapped-mainstem scenario



Top: complete real networks; dark line is this analysis's contiguous routing mainstem; larger gold entry markers denote more lateral catchment area. Middle: area-weighted lateral entry positions. Bottom: conservative same-input diagnostic, original mean-path units; curves centered separately. Each incremental catchment is counted once. Arrival units include lateral tributaries and mainstem-local inputs. Common translation overlays the original. These peaks are controlled geometry scenarios, not observed DOC events or DOC loss. Original outline representatives retained; no DOC-based selection.